KAYCEE NDUKUBA

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PROFESSIONAL SUMMARY

Applied Data Scientist (MSc) who pairs a psychology background with machine learning to predict human and customer behaviour and turn it into measurable decisions. Built a portfolio of applied machine learning and analytics projects on real-world datasets spanning customer analytics, behavioural modelling, NLP and retrieval-augmented AI, with a focus on translating technical outputs into business decisions — including a lifetime-value model concentrating ~63% of future revenue in its top predicted decile and churn models reaching ~50% of churners in the riskiest 20%. Fluent in Python, SQL and the scikit-learn / XGBoost stack, with strong statistical rigour and interpretability (SHAP).

CORE SKILLS

Machine Learning & Statistics: Predictive modelling, classification & regression, feature engineering, cross-validation, class-imbalance handling, model evaluation (ROC-AUC, precision/recall/F1), hypothesis testing, A/B thinking, EDA
Interpretability: SHAP, feature importance, coefficient analysis
Programming & Databases: Python (pandas, NumPy, scikit-learn, XGBoost), SQL (PostgreSQL, BigQuery), R, Git/GitHub
Behavioural & Domain: Behavioural statistics, RFM & customer analytics, personality/psychometrics, experimentation
NLP & AI: Sentiment analysis, text classification, LLMs, Retrieval-Augmented Generation (LangChain, FAISS)
Visualisation: Matplotlib, Seaborn, Jupyter, dashboards & stakeholder reporting

EXPERIENCE

AI & Data Scientist — Yves OS, London *Dec 2025 – Present*

- Developed behavioural analytics workflows that transformed multi-platform user-interaction data into measurable insights, used to prioritise product features and improve user experience.
- Applied large language models to generate personalised recommendations and summaries, then **evaluated outputs against defined quality criteria** to refine prompt design and improve analytical quality.
- Iteratively tested and refined AI-driven recommendation workflows using user feedback and exploratory analysis, improving recommendation relevance.
- Built engagement and productivity metrics that quantified user behaviour and informed data-driven product decisions, communicated to stakeholders through dashboards and reports.

Python · SQL · Machine Learning · LLMs · pandas · PostgreSQL · AWS

Machine Learning Engineer — Freelance / Contract *Jan 2025 – Dec 2025*

- Delivered applied machine learning and analytics across multiple private-sector engagements, translating business questions into models and measurable recommendations.
- **Selected engagements —** customer analytics, churn modelling, predictive modelling, demand forecasting and behavioural analysis.
- Built end-to-end analytical workflows — data preparation, feature engineering, model training and evaluation — in Python and SQL.
- Deployed and maintained production models with Docker on AWS/GCP, automating testing and model versioning via GitHub Actions and MLflow.

Python · scikit-learn · SQL · AWS · GCP · Docker · GitHub Actions · MLflow

Data Analyst — Plan-Net, Leicester *Oct 2023 – Nov 2024*

- Automated recurring reporting with SQL and Python, reducing manual effort and improving turnaround for business stakeholders.
- Designed and maintained ETL workflows that improved data quality, consistency and reliability across reporting pipelines.
- Partnered with cross-functional stakeholders to translate operational data into clear reports and recommendations that supported decision-making.
- Analysed infrastructure and operational data to surface trends, recurring issues and opportunities for optimisation.

SQL · Python · ETL · dbt · Snowflake · Git

SELECTED PROJECTS (FULL CODE: GITHUB.COM/KNDUKUBA17-HUB)

Customer Lifetime Value (CLV) Prediction · *Python, XGBoost, SHAP*

- Modelled 6-month customer spend from RFM behaviour on **1.07M real transactions** (UCI Online Retail II) with a leakage-safe temporal split.
- XGBoost reached test R^2 0.69 / MAE £600; the **top-10% predicted customers captured ~63% of actual future revenue** (vs 10% at random).

- Ranked the behavioural drivers of value with SHAP to make the output actionable for marketing.

Customer Churn Prediction · *Python, scikit-learn, XGBoost, SHAP*

- Predicted churn on **7,043 real telecom customers**; Logistic Regression achieved **ROC-AUC 0.842** (5-fold CV 0.845), 0.78 churn recall.
- Optimised for recall/ROC-AUC over accuracy; targeting the top-20% risk reaches **~50% of churners**.
- Handled class imbalance and explained drivers with coefficients and SHAP.

Big Five Personality Analysis (Psychology × Data Science) · *Python, PCA, K-Means*

- Validated the Big Five factor structure on **603k real survey responses** (within-trait item correlation 0.37 vs 0.01 across traits; PCA elbow at five).
- Built behavioural segmentation into four personality types with K-Means; honest read that personality is dimensional, not categorical.

Customer Churn Dashboard (interactive BI) · *JavaScript, SVG, HTML*

- Built an interactive retention dashboard on **7,043 real customers** with live client-side cross-filtering, KPI tiles (churn rate, monthly revenue at risk) and hand-built charts — no libraries.
- **Live:** kndukuba17-hub.github.io/Customer-Churn-Dashboard

EDUCATION

MSc, Applied Data Science & Cyber Security — Royal Holloway, University of London *2025 – 2026*

Modules: Machine Learning, Neural Networks, Big Data Analytics, Statistical Modelling, Cyber Threat Intelligence.

BSc, Psychology — University of Leicester *2021 – 2024*

Modules: Quantitative Research Methods, Behavioural Statistics, Cognitive Neuroscience.

A-Levels — Langley Park School *2019 – 2021*

Modules: Chemistry (A), Biology (A), Psychology (A).